MODULE 32 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

Discovery and Wave Planning

Find out what you really have, how it depends on itself, and which pieces must move together, before anything moves at all.

DOMAIN 4 of 5 EXAM WEIGHT 15% MODULE 32 of 52

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Migrations fail on what nobody knew was there. Discovery turns **assumptions** into an inventory, and dependencies into **waves**.

The discovery stack, then and now

NEED	C02-ERA TOOL	NOW
Collect inventory and dependencies	Application Discovery Service	AWS Transform discovery
Track and orchestrate the programme	Migration Hub	AWS Transform
Business case and TCO	Migration Evaluator	AWS Transform assessment
Rehost servers	Application Migration Service	AWS Transform MGN

ON THE EXAM

Migration Hub and Application Discovery Service closed to new customers on 7 November 2025.

What discovery collects

DATA	USED FOR
Server inventory and specifications	Scope, licensing, target sizing
CPU, memory, disk and network utilisation	Right-sizing, not like-for-like
Network connections, ports, processes	Dependency mapping, waves
Database metadata and metrics	Database migration path
Business owner, criticality, change windows	Wave sequencing and timing

ON THE EXAM

Collect for one to two weeks at least, so monthly and weekly jobs appear in the data.

How data is collected

AGENTLESS COLLECTOR

One appliance, many servers

- Virtual appliance in vCenter
- VM inventory and utilisation
- Connections via OS credentials
- No software on each server

IMPORTS

Data you already have

- RVTools exports
- Existing tool and CMDB data
- Fastest start
- Thinner on dependencies

THE TIEBREAKER

No agents allowed, air-gapped site: deploy the discovery appliance on-premises and upload its export.

The seven Rs

STRATEGY	MEANS	TYPICAL SIGNAL
Retire	Switch it off	No users, duplicate function
Retain	Leave it for now	Recent investment, not ready
Rehost	Lift and shift to EC2	Speed, data centre exit date
Relocate	Move the hypervisor estate	VMware estate, minimal change
Repurchase	Move to SaaS	Commodity function, licence renewal
Replatform	Lift, tinker, shift to managed	Reduce operations, little code change
Refactor	Re-architect	Scale, agility, business-critical change

ON THE EXAM

Every application gets one R. Retiring what nobody uses is the cheapest migration of all.

Building waves from dependencies

- **Group tightly coupled systems**: chatty, low-latency links move together.
- **Loose links can span waves** through hybrid connectivity.
- **Pilot wave first**: low risk, few dependencies, representative patterns.
- **Respect business calendars**: freezes, peaks, change windows.
- **Shared services early**: identity, DNS, monitoring ready before workloads.

> ON THE EXAM
>
> Splitting a chatty application from its database across waves turns LAN latency into WAN latency.

A migration programme, end to end

01
Discover
Inventory, utilisation, connections

02
Assess
Seven Rs, business case

03
Plan
Application groups and waves

04
Prepare
Landing zone, connectivity, shared services

05
Migrate
Pilot, then waves, then cutovers

ON THE EXAM

The landing zone and connectivity come before the first wave, not during it.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Starting discovery in 2026	AWS Transform, not Migration Hub
Agents not allowed, air-gapped	Discovery appliance, export uploaded
App and database talk constantly	Same wave
Unused for months	Retire
Commodity function, licence renewal	Repurchase
Choosing the first wave	Low-risk, representative pilot

ON THE EXAM

If an answer names Application Discovery Service or Migration Hub for a new programme, it is stale.

> Meridian Freight is starting a new data centre exit in 2026 covering 1,200 VMware virtual machines. It has never used AWS migration tooling before.
>
> Leadership wants an inventory, utilisation-based sizing, a dependency map, proposed waves and a business case.

What should the solutions architect use?

A AWS Application Discovery Service agents with AWS Migration Hub

B AWS Transform, with its discovery tool and assessment

C A spreadsheet survey of application owners

D AWS Transform MGN replication agents on every server

◆ PAUSE AND DECIDE

What should the solutions architect use?

A ADS with Migration Hub

Closed to new customers since November 2025.

B AWS Transform

Discovery, dependencies, waves and business case in one service.

C Spreadsheet survey

Opinions, not measured data; misses dependencies.

D MGN agents

Replicates servers for migration; not a discovery tool.

THE SIGNAL

New customer, new programme, discovery to waves: AWS Transform.

> Meridian Freight's dependency map shows the billing application making thousands of small queries per transaction to an Oracle database. The database is also read nightly by a reporting system.
>
> The team is assigning these three systems to waves.

How should the solutions architect plan the waves?

A Billing first, then the database a month later

B The database first, then billing and reporting later

C Billing and its database in the same wave; reporting in a later wave over hybrid connectivity

D All three in separate waves, by business owner

◆ PAUSE AND DECIDE

How should the solutions architect plan the waves?

A **Billing first**

Thousands of queries per transaction would cross the WAN for a month.

B **Database first**

Same problem in reverse for billing.

C **Chatty pair together**

Keeps the latency-sensitive link local; batch reads tolerate WAN.

D **By business owner**

Ignores the technical dependency entirely.

THE SIGNAL

Chatty and latency-sensitive: same wave. Batch and tolerant: can span waves.

> During assessment, Meridian Freight finds a self-hosted expenses application on four servers. Its licence and support contract expire in six months, and several SaaS products provide the same function.
>
> No custom features are required.

Which migration strategy fits this application?

A Repurchase: move to a SaaS expenses product

B Rehost the four servers to EC2

C Refactor into microservices on AWS

D Retain on-premises indefinitely

◆ PAUSE AND DECIDE

Which migration strategy fits this application?

A Repurchase

Commodity function, expiring licence, no customisation: buy it as SaaS.

B Rehost

Moves servers whose licence is about to lapse.

C Refactor

Rebuilding a commodity function is wasted effort.

D Retain

Blocks the data centre exit.

THE SIGNAL

Commodity, licence renewal due, nothing custom: repurchase.

> Meridian Freight has 40 planned waves. The programme lead must choose the first wave to prove the migration process, tooling and runbooks.
>
> The board is watching the first wave closely.

Which applications should form the first wave?

A The customer booking platform, to show value quickly

B The largest applications, to reduce the remaining scope

C A random sample from every business unit

D A few low-risk applications with few dependencies that use the common technology patterns

◆ PAUSE AND DECIDE

Which applications should form the first wave?

A Booking platform first

Highest risk while the process is still unproven.

B Largest first

Big and complex before the team has learned anything.

C Random sample

Ignores risk and dependencies.

D Low-risk, representative pilot

Proves the factory safely on patterns that recur later.

THE SIGNAL

First wave: low risk, few dependencies, representative of what follows.

> A Meridian Freight data centre in a regulated subsidiary has no internet access, and its security policy forbids installing software on production servers.
>
> The team still needs inventory, utilisation and network dependency data for assessment.

How should the team collect discovery data?

A Install discovery agents on each server

B Ask application owners to document dependencies

C Deploy the AWS Transform discovery tool appliance in vCenter, collect offline, and upload the export when approved

D Start replicating servers with AWS Transform MGN and observe traffic in AWS

◆ PAUSE AND DECIDE

How should the team collect discovery data?

A Agents per server

Forbidden by the security policy.

B Owner documentation

Incomplete and unmeasured.

C Discovery appliance, offline

No per-server software; works air-gapped; export uploaded later.

D Replicate first

Installs replication agents, and migrates before assessing.

THE SIGNAL

No agents, no internet: the on-premises discovery appliance and an offline export.

> Meridian Freight's discovery data shows 140 servers with near-zero CPU and network activity for six months. Owners for 120 of them confirm the systems were replaced years ago.
>
> The programme must cut migration scope and cost.

What should the solutions architect recommend for those 120 servers?

A Rehost them in the last wave

B Retire them, after confirming no hidden dependencies and archiving data as required

C Retain them on-premises

D Replatform them to managed services

◆ PAUSE AND DECIDE

What should the solutions architect recommend for those 120 servers?

A **Rehost later**

Pays to move systems nobody uses.

B **Retire**

Removes scope and cost; checks and archiving cover the risk.

C **Retain**

Keeps paying for idle servers, and blocks the exit.

D **Replatform**

Effort spent modernising dead systems.

THE SIGNAL

Measured idle, owners confirm replaced: retire, with care.

What to carry out of this module

- **Measure, do not ask**: inventory, utilisation and connections over weeks.
- **AWS Transform** for new programmes; Migration Hub and ADS are closed to new customers.
- **One of seven Rs** per application; retire whatever you can.
- **Chatty dependencies move together**; loose ones can span waves.
- **Pilot first**: low risk, representative, after the landing zone is ready.

> ON THE EXAM
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> Discover, assess, plan, prepare, then migrate.